WORKED SOLUTION

STEP III 2003, Question 7

Pure • Simson Line

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q7

The key idea

The foot of a perpendicular is found by walking from P along the normal to the line until we hit it, so write $N = P + \lambda(\text{normal})$ and solve for λ. For the second part, the condition on s and t is a product of two gradients equal to -1: it says that PA and PB are perpendicular. Turned into a circle equation, it is exactly what makes the collinearity check collapse.

Solution

The coordinates of N

The line AB has equation $\dfrac{x}{a} + \dfrac{y}{b} = 1$, that is

$$bx + ay = ab.$$

A normal vector to this line is (b, a), so N, the foot of the perpendicular from P, has the form

$$N = (s + \lambda b,\ t + \lambda a)$$

for some λ. Since N lies on AB,

$$b(s + \lambda b) + a(t + \lambda a) = ab \quad \Longrightarrow \quad \lambda = \frac{ab - bs - at}{a^2 + b^2}.$$

Hence

$$x = s + \lambda b = \frac{s(a^2 + b^2) + ab^2 - b^2 s - abt}{a^2 + b^2} = \frac{ab^2 - abt + a^2 s}{a^2 + b^2} = \frac{ab^2 - a(bt - as)}{a^2 + b^2},$$

$$y = t + \lambda a = \frac{t(a^2 + b^2) + a^2 b - abs - a^2 t}{a^2 + b^2} = \frac{a^2 b + b^2 t - abs}{a^2 + b^2} = \frac{a^2 b + b(bt - as)}{a^2 + b^2},$$

as required.

N lies on XY

First, what the condition says. For it to make sense we need $s \neq 0$ and $s \neq a$, and since the product is -1 rather than 0, also $t \neq 0$ and $t \neq b$. Multiplying out,

$$t(t-b) = -s(s-a) \quad \Longleftrightarrow \quad s^2 + t^2 = as + bt. \qquad (*)$$

Now $X = (s, 0)$ and $Y = (0, t)$. As $s \neq 0$ and $t \neq 0$, these are distinct points off the origin, and the line XY is

$$\frac{x}{s} + \frac{y}{t} = 1, \qquad \text{that is} \qquad tx + sy = st.$$

So we must show that $tx + sy = st$ for the coordinates of N.

Ask yourself Should we substitute and hope, or is there a tidy way to organise the algebra?

Write $k = bt - as$ to keep things short. Using the formulae above,

$$\begin{aligned}(a^2 + b^2)(tx + sy) &= t(ab^2 - ak) + s(a^2 b + bk) \\ &= ab^2 t + a^2 bs + k(bs - at).\end{aligned}$$

The last product expands neatly:

$$k(bs - at) = (bt - as)(bs - at) = b^2 st - abt^2 - abs^2 + a^2 st = (a^2 + b^2)st - ab(s^2 + t^2).$$

Therefore

$$(a^2 + b^2)(tx + sy) = (a^2 + b^2)st + ab(as + bt - s^2 - t^2).$$

This holds for every P. By $(*)$ the bracket is zero, so $tx + sy = st$ and N lies on the line XY.

Geometrical interpretation

The gradient of PB is $\dfrac{t-b}{s}$ and the gradient of PA is $\dfrac{t}{s-a}$, so the condition says that $PA \perp PB$: the angle APB is a right angle. Equivalently, $(*)$ is

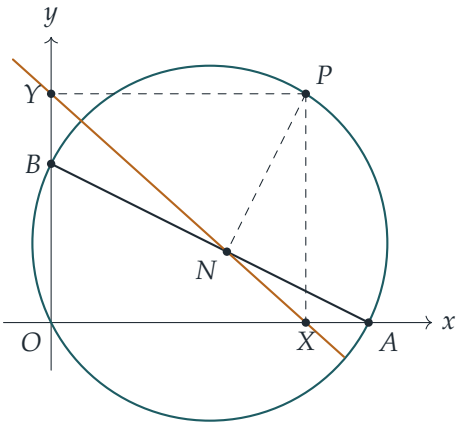
$$\left(s - \tfrac{a}{2}\right)^2 + \left(t - \tfrac{b}{2}\right)^2 = \frac{a^2 + b^2}{4},$$

so P lies on the circle with diameter AB. Since the angle AOB is also a right angle, this circle passes through O: it is the *circumcircle of triangle OAB*.

The points X, Y and N are the feet of the perpendiculars from P to the three sides OA, OB and AB of the triangle. So the result says:

If P lies on the circumcircle of a triangle, the feet of the perpendiculars from P to the three sides (extended if necessary) are collinear.

This is the *Simson line* theorem, here proved for a right-angled triangle placed with its right angle at the origin.



With $a = 4$, $b = 2$: P on the circle through O, A, B; the line XY (amber) passes through N.

Checks

- Put $P = O$, i.e. $s = t = 0$: the formula gives $N = \left(\frac{ab^2}{a^2+b^2}, \frac{a^2b}{a^2+b^2}\right)$, which lies on $bx + ay = ab$ and on the line $y = \frac{a}{b}x$ through O perpendicular to AB, as it should.
- With $a = 4$, $b = 2$ and $P = (2 + \sqrt{5}\cos 1,\ 1 + \sqrt{5}\sin 1) \approx (3.208, 2.882)$ on the circle, the formula gives $N \approx (2.214, 0.893)$, and $\frac{2.214}{3.208} + \frac{0.893}{2.882} \approx 0.690 + 0.310 = 1$, so N is on XY.
- The identity $(a^2 + b^2)(tx + sy - st) = ab(as + bt - s^2 - t^2)$ also gives the converse: for P off the axes, N is on XY *only* if P is on the circle.

Common slips

- Using (a, b) as the normal to AB. The direction of AB is $(-a, b)$, so the normal is (b, a).
- Expanding everything at once and drowning. Grouping $bt - as$ as one symbol, as the given answer suggests, keeps the algebra to a few lines.
- Stopping at "angle APB is a right angle". The interpretation the question is after is the Simson line: X, Y, N are feet of perpendiculars to the three sides of triangle OAB.

Beyond STEP

The Simson line holds for any triangle, and a synthetic proof uses cyclic quadrilaterals and equal angles in the same segment. Its converse is also true, and as P moves round the circumcircle the Simson lines envelope a deltoid (Steiner's hypocycloid). Coordinate

and complex-number methods make such results mechanical; Needham, *Visual Complex Analysis*, shows how complex numbers handle circle geometry elegantly, and Coxeter and Greitzer, *Geometry Revisited*, covers the Simson line and much more classical triangle geometry.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.